

DOCUMENT 2: SOLUTION REQUIREMENTS

Functional Requirements

S.NO	REQUIREMENT CATEGORY	REQUIREMENT DESCRIPTION	PRIORITY
1	Authentication	No user authentication needed for MVP. Users can generate comics without login. Optional JWT-based authentication for future features like saving history	Low
2	Authorization Levels	Single user role - no admin/user distinction. Everyone has equal access to all features. No role-based access control needed	Low
3	External Interfaces	Google Gemini API (Flash 1.5 for outlines, Pro 1.5 for story/narration), Hugging Face Inference API / Diffusers (Stable Diffusion v1.5 / FLUX.1-schnell for image generation)	High
4	Transaction Processing	User submits form → FastAPI validates input (Pydantic) → Gemini Flash generates 5-panel JSON outline → Gemini Pro expands with narration/dialogue → Image pipeline generates 5 panel illustrations → Layout builder assembles images + text → PDF exporter compiles → Preview page renders → User downloads PDF	High
5	Reporting	Comic preview page with panel-by-panel view (title, scene description, narration caption, dialogue, illustration). Multi-page downloadable PDF with title page and all 5 panels	Medium
6	Business Rules	Must accept: prompt (text, max 2000 chars), character name, setting (forest/space/city/beach/castle + custom), tone (funny/dramatic/adventurous/romantic/mysterious), art style (anime/cartoon/realistic/comic). Default 5 panels. Output includes per-panel: title, scene, narration, dialogue, illustration	High

7	Compliance	No user data stored persistently (GDPR-friendly). API keys stored server-side in .env file, never exposed to client. No cookies, no tracking, no user profiling	Medium
8	Other (Fallback)	Intelligent mock fallback mode when no API keys detected - generates placeholder content with sample images/text for UI/PDF testing without AI calls	Medium

Non-Functional Requirements

S.NO	NFR CATEGORY	REQUIREMENT DESCRIPTION	TARGET METRIC
1	Performance & Speed	Story generation (Flash + Pro) completes in < 5 seconds. Page load < 2 seconds. Image generation depends on GPU/API speed	Page loads < 2s; Story gen < 5s
2	Scalability	Stateless FastAPI architecture. Images stored temporarily in static/panels/. Async support for concurrent requests	Supports concurrent user requests
3	Security & Data Privacy	API keys stored in .env, never client-side. Input sanitization. No persistent user data. Rate limiting for production	Keys encrypted at rest
4	Reliability & Availability	Works with or without API keys (mock fallback). Graceful error handling with user-friendly messages. Try/except on all API calls	99% uptime local dev
5	Usability & Accessibility	Clean dark glassmorphism UI, responsive design (mobile/tablet/desktop), clear error messages, interactive Swagger docs at /docs	WCAG basic level compliance
6	Maintainability	Modular architecture - separate files for each concern. Well-commented code. Easy to extend with new features	Modular, documented codebase